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Product Feature

1. 15-bit absolute encoder, one-turn pulse up to 32768.
2. Multi-stage DD motor structure, large torque output.
3. Integrated servo, simplified wiring, ultra-small size.
4. Low noise, low vibration, high-speed positioning, high reliability.
5. FOC field oriented vector control, supports position/speed closed loop.
6. Can work in zero lag given pulse state and follow zero lag.
7. 16-bit electronic gear function. 8.
- Modbus RTU communication / canopen communication (choose one of the two) 9. Position mode, support pulse + direction signal
10. With stall protection, over-current protection, over-voltage protection.



CANopen / 485 isolated communication Built-in isolated power supply
 Low power consumption Multi-turn absolute value automatic charging Power failure
 Alarm Pulse + direction control 1500/2500RPM speed optional
 Integrated servo built-in drive, simplified system

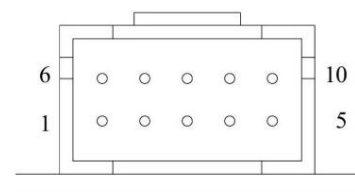
Parameter T able

Model parameters		60AIM40F	60AIM40FH
power supply	Voltage	36VDC±10%	36VDC±10%
	Current	7A	7A
Motor parameters	Torque	2NM	1NM
	Rated Speed	1000RPM	2000RPM
	Maximum Speed	1500RPM	2500RPM
	Power	200W	200W
	Resistance	0.53ÿ	0.53ÿ
	Inductance	0.5mH	0.5mH
	Moment of Inertia	1.74×10^{-4} KG / M^2	1.74×10^{-4} KG / M^2
	Feedback signal		Single-turn 15-bit magnetic encoder (32768 pulses per turn)
Cooling method		Natural cooling	
Weight			
Position control mode maximum input pulse frequency 500KHz			
	Pulse command mode	Pulse+direction, phase A+phase B	
	Electronic gear ratio	Setting range 1~65535 Ratio 1~65535	
	position sampling frequency	2KHz	
Protection function		Stall alarm	
communication		RS485 modbus RTU communication / canopen communication (choose one)	
interface usage environment	Ambient temperature	0~40°	
	The maximum allowable temperature of the motor is 85°		
	humidity	5~95%	

Interface Definition

Terminal number: Facing the terminal, the left side is first.

Terminal number	name	Function
1	+24V	The positive pole of the DC power supply is +24V. Connecting the positive and negative poles in reverse will directly short-circuit the power supply and may also damage the driver.
2	GND	DC power ground. Reverse connection of positive and negative will directly short-circuit the power supply and may also damage the driver
3	PU+ (+5V) pulse control signal:	the rising edge of the pulse is valid; PU- is 3.3~5V when high level, and 0~0.5V when low level.
4	PU- (PU)	In order to reliably respond to pulse signals, the pulse width should be greater than 1.2 μ s. If +12V or +24V is used, a series resistor is required.
5	DIR+ (+5V) Direction signal:	high/low level signal. To ensure reliable commutation of the motor, the direction signal should precede the pulse signal.
6	DIR- (DIR)	takes at least 5 μ s to establish. DIR- is 3.3~5V when high level, and 0~0.5V when low level.



Terminal serial number: Facing the terminals, the lower row from left to right are 12345, and the upper row from left to right are 6 7 8 9 10.

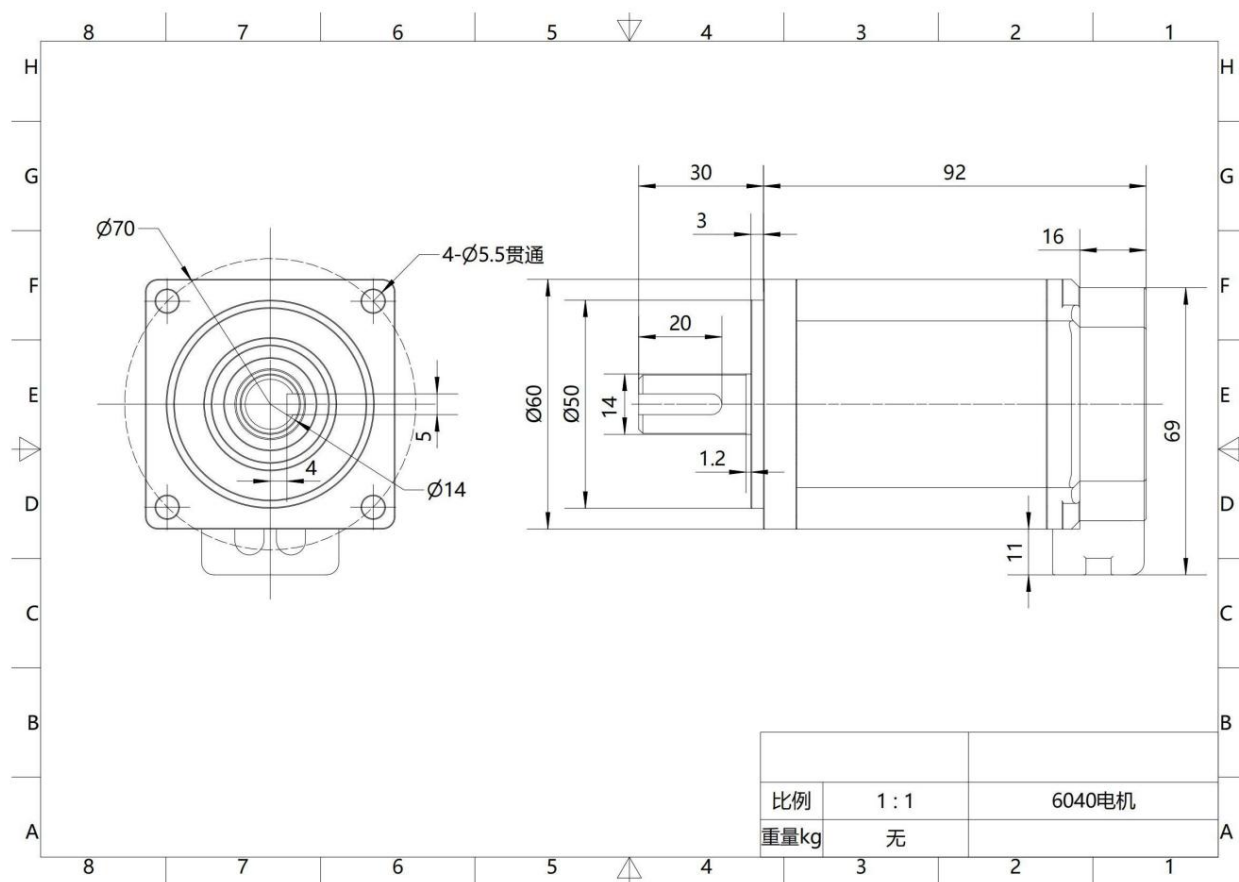
The 485 version plug is defined as follows:

Terminal No.	Name	Function
1	NC	
2	485A	485 Communication positive terminal
3	485B	485 Communication negative terminal
4	EN+ Enable signal positive	
5	EN- Enable signal negative	
6	COM output signal and 485 power common ground.	
7	WR alarm signal output, internal is optocoupler NPN output. Normally it is high impedance state, and it is connected to COM when alarm occurs.	
8	RDY/PF Servo ready signal/in-position signal. After power-on, there is a signal (conduction). When the following error is less than 0.5°, There is signal (conduction), the following error is greater than 0.5°, and there is no signal (high impedance state).	
9	ZO Encoder zero point output. When there is a zero point signal, the optocoupler NPN outputs a conduction signal.	
10	485_5V 485 communication 5V power supply, requires external power supply. (This power supply is supplied by the controller)	

The CAN version plug is defined as follows:

Terminal No.	Name	Function
1	CANL	CAN communication port. To use CAN communication, CAN_5V,COM needs to be powered with 5V.
2	NC	
3	NC	
4	CANH	CAN communication port. To use CAN communication, CAN_5V,COM needs to be powered with 5V.
5	NC	
6	COM Output signal and CAN power common ground.	
7	WR alarm signal output, internal is optocoupler NPN output. Normally it is high impedance state, and it is connected to COM when alarm occurs.	
8	RDY/PF Servo ready signal/in-position signal. After power-on, there is a signal (conduction). When the following error is less than 0.5°, There is signal (conduction), the following error is greater than 0.5°, and there is no signal (high impedance state).	
9	ZO Encoder zero point output. When there is a zero point signal, the optocoupler NPN outputs a conduction signal.	
10	CAN_5V CAN communication 5V power supply, requires external power supply. (This power supply is supplied by the controller)	

Note: 60AIM40F



Note: 60AIM40BF (brake)

